

Activity 4

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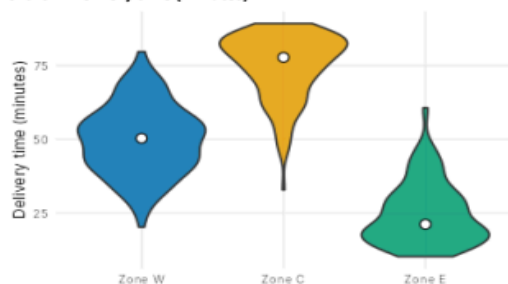
Course Code : DSA0610 Data Handling And Visualization

DSA06 - Data Handling and Visualization | Activity: Identify Skewness using Violin Plots | Slot C

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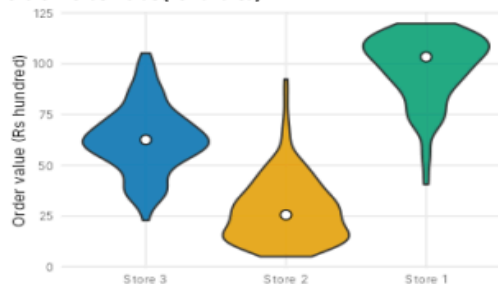
Part A (18 marks). Each chart is a violin plot (white dot = median) made in R with ggplot2. For every violin circle the skew (1 mark) and the relation of the mean to the median (0.5 mark).

Chart 1: Delivery time (minutes)



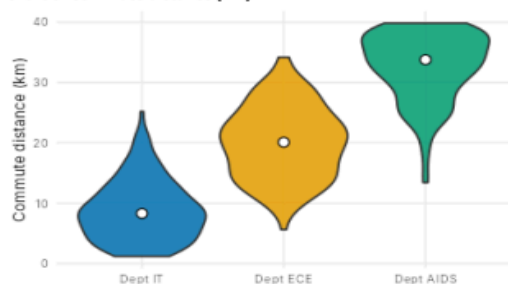
Violin	Skew (circle one)	Mean vs Median (circle)
Zone W	Right / Left / Symm.	Mean > / < / ~ Median
Zone C	Right / Left / Symm.	Mean > / < / ~ Median
Zone E	Right / Left / Symm.	Mean > / < / ~ Median

Chart 2: Order value (Rs hundred)



Violin	Skew (circle one)	Mean vs Median (circle)
Store 3	Right / Left / Symm.	Mean > / < / ~ Median
Store 2	Right / Left / Symm.	Mean > / < / ~ Median
Store 1	Right / Left / Symm.	Mean > / < / ~ Median

Chart 3: Commute distance (km)



Violin	Skew (circle one)	Mean vs Median (circle)
Dept IT	Right / Left / Symm.	Mean > / < / ~ Median
Dept ECE	Right / Left / Symm.	Mean > / < / ~ Median
Dept AIDS	Right / Left / Symm.	Mean > / < / ~ Median

Chart 4: Server response (ms)



Violin	Skew (circle one)	Mean vs Median (circle)
Server 2	Right / Left / Symm.	Mean > / < / ~ Median
Server 3	Right / Left / Symm.	Mean > / < / ~ Median
Server 1	Right / Left / Symm.	Mean > / < / ~ Median

Part B (2 marks)

B1. In Chart 1, violin "Zone C" is not symmetric. Towards which side is its tail, and which is the better measure of a typical delivery time here: mean or median? Give one reason. [1]

B2. Write one ggplot2 line that draws violins of value for each group from a data frame df (columns group, value), with the median shown. [1]

Chart 1: Delivery time (minutes)

Violin	Skew	Mean vs Median
Zone W	Right / Left / Sym.	Mean > / < / ~ Median
Zone C	Right / Left / Sym.	Mean > / < / ~ Median
Zone E	Right / Left / Sym.	Mean > / < / ~ Median

Chart 2: Order value (Rs hundred)

Violin	Skew	Mean vs Median
Store 3	Right / Left / Sym.	Mean > / < / ~ Median
Store 2	Right / Left / Sym	Mean > / < / ~ Median
Store 1	Right / Left / Sym	Mean > / < / ~ Median

Chart 3: Commute distance (km)

Violin	Skew	Mean vs Median
Dept IT	Right / Left / Sym.	Mean > / < / ~ Median
Dept ECE	Right / Left / Sym	Mean > / < / ~ Median
Dept AIDS	Right / Left / Sym	Mean > / < / ~ Median

Chart 4: Server response (ms)

Violin	Skew	Mean vs Median
Server 2	Right / Left / Sym.	Mean > / < / ~ Median
Server 3	Right / Left / Sym.	Mean > / < / ~ Median
Server 1	Right / Left / Sym.	Mean > / < / ~ Median

Part B – 2 Marks

B1. Chart 1 – Zone C

Question:

In Chart 1, violin “Zone C” is not symmetric. Towards which side is its tail, and which is the better measure of a typical delivery time here: mean or median? Give one reason.

Answer:

- Tail: Left side
- Better measure: Median
- Reason: The distribution is left-skewed, so the median is less affected by the long tail/outliers than the mean.

B2. ggplot2 code

Question:

Write one ggplot2 line that draws violins of **value** for each group from a data frame (**group**, **value**), with the median shown.

Answer:

```
ggplot(df, aes(x = group, y = value)) +  
  geom_violin() +  
  stat_summary(fun = median, geom = "point")
```